



P23

Project-Level Five-Dimensional Evaluation: From Work Orders to a Project Failure Taxonomy

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Project-Level Five-Dimensional Evaluation: From Work Orders to a Project Failure Taxonomy

Working Title (EN): Project-Level Five-Dimensional Evaluation: Aggregating UDOS Work-Order Metrics into EPOB-Style Project Quality and a Failure Taxonomy

In-volume engineering paper P23 • raising UDOS's measurement granularity from "work order" to "project" • borrowing the EPOB five dimensions and failure-distribution diagnosis • UDOS Reasoning Engine v7.6.0 • Fang Wenxin • 2026-09-21

One sentence first: UDOS's current measurement unit is "whether one work order got done," but what a multi-Agent system actually delivers is "one project"—how good the plan is, how accurately tasks are assigned, how smoothly handoffs go, how complete the deliverables are, how fast and costly it runs: five things work-order success rate cannot answer. EPOB (OpenReview NtPIgzPtOU) raises multi-Agent system evaluation from "single-task success rate" to the **project lifecycle level**, and stresses looking at the **failure**

distribution rather than only the success rate. This paper connects this five-dimensional framework into UDOS, giving the logic of how work-order metrics aggregate into project metrics—a **design proposal • to implement**.

Evidence caliber (stated only once in the whole paper): this paper is an evaluation-methodology design paper, **containing no new UDOS measured numbers**. The five dimension names and failure taxonomy categories of the external paper EPOB (End-to-End Project Orchestration Benchmark, OpenReview NtPlgzPtOU, 2026-05) have been verified; concrete constructions such as its "180-unit robustness package" are paper reports, not independently reviewed. UDOS old metrics (rework rate, message fan-in, completion rate, handoff completeness, throughput curves) all point back to v7.5.0 papers, evidence grades marked in text. This paper's work-order→project aggregation logic is a **design proposal • to implement**, aggregation weight calibration [RESULT NEEDED] .

Abstract

Multi-Agent system evaluation long stayed at "single-task success rate": run one task, done records 1, failed records 0, finally averaged. This granularity has two ills—it only tells you "good or bad," not "where broken"; it only measures "one work order," cannot measure "one project." EPOB (OpenReview NtPlgzPtOU, 2026) raises the evaluation unit to the project lifecycle, measuring along five dimensions: Plan Quality, Assignment Quality, Coordination, Deliverable Quality, Efficiency; more importantly it stresses the **failure distribution rather than only the success rate**, using five failure labels—planning / delegation / handoff / review / recovery—to answer "which link broke." UDOS v7.5.0's measurement actually already covered the "parts" of these five dimensions—P1 measures fan-in and rounds (coordination), P2 measures stopping decisions (coordination/recovery), P3 measures rework and mis-isolation (recovery), P10 measures handoff completeness (handoff/deliverable), P9 measures throughput curves (efficiency)—but they are all pinned at the "single work order" granularity, not aggregated into project-level scores. This paper does two things. First, mapping UDOS's existing work-order metrics one by one onto the EPOB five dimensions, marking which dimensions already have instruments and which lack them. Second, giving the two-layer work-order→project aggregation logic: the bottom layer remains single-work-order metrics, the middle layer classifies the same project's work orders by failure label, the top layer uses project-level weighting to obtain five-dimensional scores. This paper also points out EPOB's drawback: it is a "framework-centric" evaluation, measuring "whether this framework executes projects well," not "whether this framework matches this task"—the latter being precisely the unique value of UDOS P1's topology selection decision tree, not replaceable by EPOB. This paper is methodology design, aggregation weights and thresholds pending [RESULT NEEDED] .

Keywords: project orchestration; evaluation benchmark; failure taxonomy; five-dimensional metrics; work-order aggregation; EPOB; UDOS P1/P3/P9/P10

Structured Abstract (background problem → method/argument → evidence → contribution)

- **Background problem:** UDOS's metrics are all at work-order granularity; single-work-order success rate cannot answer "whether one project is well planned, accurately assigned"; looking only at success rate also cannot answer "which link broke."
- **Core argument:** the evaluation unit should rise from work order to project; using the EPOB five dimensions as the skeleton, using the five failure labels planning/delegation/handoff/review/

recovery for diagnosis, aggregating UDOS's existing work-order metrics in two layers into project-level five-dimensional scores.

- **Evidence clues:** (1) EPOB five dimensions and failure taxonomy (OpenReview NtPlgzPtOU, existence verified); (2) UDOS existing work-order metrics (P1 fan-in/rounds, P2 stopping decisions, P3 rework/mis-isolation, P10 handoff completeness, P9 throughput, pointing back to papers); (3) the known boundary that EPOB does not evaluate "framework-task matching."
 - **Contribution:** a mapping table from UDOS work-order metrics to the EPOB five dimensions; two-layer aggregation logic; a five-dimension instrument gap list; the demarcation between EPOB's drawback and UDOS's unique value.
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1 Problem: Work-Order Success Rate Cannot Answer Project Questions

1.1 The Two Ills of Single-Task Success Rate

The vast majority of UDOS v7.5.0 measurements—P1's fan-in/rounds, P2's stopping decision correctness, P3's rework rate, P10's handoff completeness—are **single-work-order** granularity. Run one work order, see whether it got done. This granularity is correct during system debugging, but it has two ills:

- **Only says good or bad, not where broken.** A project failed—was the plan wrong from the start (planning)? Or the plan right but the wrong person assigned (delegation)? Or information lost at handoff (handoff)? Single-work-order success rate compresses these five failures into the same 0, diagnostic value zero.
- **Only measures one work order, cannot measure one project.** Real delivery is "one project": multiple work orders, multiple handoff rounds, with closing. Single-work-order average success rate may be high, while project-level plan quality and delivery completeness are a mess.

1.2 EPOB's Answer: Project Lifecycle + Failure Distribution

EPOB (End-to-End Project Orchestration Benchmark, OpenReview NtPlgzPtOU, 2026-05) raises the evaluation unit from single task to the **project lifecycle**, scoring along five dimensions:

- **Plan Quality:** how well the task is decomposed;
- **Assignment Quality:** to whom tasks are assigned, how accurately;
- **Coordination:** fan-in, rounds, stopping decisions;
- **Deliverable Quality:** completion rate, handoff completeness;
- **Efficiency:** throughput, cost curves.

Its more crucial methodological claim is the **failure distribution rather than only the success rate**: attaching taxonomy labels to each failure—planning (wrong at the planning stage), delegation (wrong assignment), handoff (information lost at handoff), review (review missed it), recovery (recovery failed)—turning evaluation from "good or bad" into "where broken."

1.3 UDOS Does Not Start from Zero

Interestingly, the "parts" of UDOS's five dimensions are already scattered across papers. This paper's work is not building new instruments but **aggregating the scattered work-order metrics into project-level five-dimensional scores**.



Figure 1 Left column UDOS existing work-order metrics, middle column EPOB five dimensions, right column the instrument status of the five dimensions in UDOS (existing/to supplement). Conceptual illustration, not measured data.

2 Five-Dimension Mapping and Instrument Gaps

2.1 The Mapping Table

Table 1 Mapping from UDOS work-order metrics to the EPOB five dimensions

EPOB dimension	Meaning	UDOS corresponding work-order metric	Instrument status
Plan Quality	task decomposition quality	P1 topology selection, P3 governance design decomposition reasonableness	partial: topology comparison exists, lacks direct "decomposition quality" metric
Assignment Quality	task assignment quality	P11 internal market bidding, P17 scope granting	partial: bidding mechanism exists, lacks "assignment right/wrong" label
Coordination	coordination quality	P1 fan-in/rounds, P2 stopping decisions	existing: protocol-layer measurement mature
Deliverable Quality	deliverable quality	P3 completion rate, P10 handoff completeness	existing: contract tests cover
Efficiency	efficiency	P9 throughput curves, cost	existing: cpu-proto curves

2.2 Where the Gaps Are

The mapping table exposes two clear gaps:

- **Plan Quality lacks a direct metric.** UDOS can measure "the layered tree presses fan-in to a constant" (P1, verified extrapolation), but this is "coordination efficiency," not "whether this

project's task decomposition itself is reasonable." A poorly decomposed plan can still be efficiently executed by the layered tree.

- **Assignment Quality lacks an "assignment right/wrong" label.** P11 has a bidding mechanism, but no post-hoc marking of "whether assigning this to X was optimal." Without this label, Assignment Quality can only compute "how fast assigned," not "whether assigned correctly."

These two gaps are precisely what this paper fills: they are not new systems, but on existing work-order records **adding two post-hoc labels**—decomposition reasonableness label, assignment right/wrong label.

3 The Two-Layer Work-Order→Project Aggregation Logic (design proposal • to implement)

3.1 Why Cannot Directly Average

Directly averaging the success rates of all work orders in a project is the practice EPOB criticizes—it lumps planning failure and recovery failure together. This paper adopts two-layer aggregation:

Bottom layer (work-order level): each work order records a set of atomic metrics—whether completed, failure label (one of five), fan-in, rounds, handoff completeness, cost.

Middle layer (failure distribution): classifying the same project's work orders by failure label, obtaining the count distribution of five failure classes $n_p, n_d, n_h, n_r, n_{\text{rec}}$. This layer answers "which link broke."

Top layer (five-dimensional scores): mapping the middle-layer distribution to five-dimensional scores. For example—

$$\text{Plan Quality} \propto 1 - \frac{n_p}{\sum n} \tag{1}$$

i.e. the higher the planning failure share, the lower Plan Quality; Assignment Quality inversely related to n_d , Coordination related to rounds/fan-in, Deliverable Quality related to completion rate/handoff completeness, Efficiency related to unit delivery cost. The normalization weights of each dimension are design parameters, pending [RESULT NEEDED] calibration.

3.2 Granularity Difference from EPOB

Must honestly point out: EPOB's evaluation granularity is "project," UDOS's current measured granularity is "work order." This paper gives **the logic of aggregating from work order to project**, not project-level scores already run out. If UDOS is to submit to a systems conference, this aggregation must run through on real multi-work-order projects—otherwise P23 is only methodology, no data. This is precisely the core of [RESULT NEEDED].



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Figure 2 Work-order-level atomic metrics → five-class failure distribution → project-level five-dimensional scores; diagnostic value concentrated in the middle failure-distribution layer. Conceptual illustration.

4 Drawbacks / Failure Boundaries / Falsifiable Conditions

4.1 Drawbacks (restraint that must be kept when citing EPOB)

1. **EPOB does not evaluate "framework-task matching."** It measures "whether this framework's project execution effect is good," **not "when to use star topology, when to use layered."** This is not a drawback EPOB hides, but its problem set does not contain this question. UDOS P1's topology selection decision tree precisely fills this gap—do not take EPOB as "we have been externally evaluated"; it and UDOS are complementary, not duplicate.
2. **Failure taxonomy labels have a human-judgment component.** The boundary between planning and delegation is not clean in ambiguous cases; label consistency needs annotation specifications, otherwise the failure distribution itself has noise.
3. **Framework-centric ≠ task-centric.** EPOB takes the framework as the object under test; UDOS's interest is "structural prior determines the capability upper limit," the two ask from different angles; when transferring its five dimensions be aware of this perspective difference.

4.2 Failure Boundaries

- If a project's work orders are highly isomorphic (P9's homogeneous copies), the failure distribution concentrates on a single label, five-dimension discrimination drops—this is not the aggregation logic being wrong, but the project itself lacking diversity; EPOB-style five dimensions are inherently insensitive to homogeneous projects.
- If the two new labels Plan Quality / Assignment Quality rely on LLM post-hoc judging, they introduce the LLM-as-judge bias P12 already discussed; these two labels best also go through execution-style determination (e.g. comparison with oracle decomposition/assignment), rather than purely subjective scoring.

4.3 Falsifiable Conditions

- **F23-a:** if the project-level five-dimensional scores obtained by two-layer aggregation are **not correlated** with domain experts' qualitative ranking of the same batch of projects (rank correlation below the design threshold), then aggregation weight calibration fails, the five-dimension mapping needs redoing.
 - **F23-b:** if the failure distribution diagnosis ("broke at handoff") has a high inconsistency rate with post-hoc root cause analysis conclusions, then the taxonomy label specification needs revision.
 - The above are post-hoc-scorable criteria under a design proposal, not established conclusions; weights and thresholds pending [RESULT NEEDED].
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5 Conclusion

EPOB raises evaluation from "single-task success rate" to "project lifecycle + failure distribution," the direction correct. What UDOS must do is not rebuild a benchmark, but aggregate its work-order metrics scattered in P1/P2/P3/P9/P10/P11 according to the EPOB five dimensions in two layers into project-level scores, and fill the two gap labels Plan Quality, Assignment Quality. At the same time must stay clear-headed: EPOB does not answer "whether the framework matches the task," that being UDOS P1's topology selection tree's exclusive ground—the five-dimension evaluation is a checkup form, not a matching plan.

Work-order success rate only tells you whether it got done; the failure distribution tells you which link broke. Evaluating a system, first measure where it breaks, then measure whether it is good.

References

Evaluation methodology foundations

1. Dean, J., & Bowen, K. (1987). Calibration of Acceptance Measures for Software Defects (early thinking on defect classification and density analysis).
2. Henard, R. H., & Manner, M. K. (2014). Predicting product success with a quantitative model. (general method for quality multi-dimensional evaluation)

2026 frontier material (existence verified online 2026-09-21; concrete construction numbers are paper reports, not independently reviewed)

1. EPOB (2026). End-to-End Project Orchestration Benchmark. OpenReview NtPIgzPtOU. (Plan/Assignment/Coordination/Deliverable/Efficiency five dimensions; planning/delegation/handoff/review/recovery failure taxonomy)

UDOS paper volumes (this series, evidence grades marked in text)

1. UDOS v7.5.0 paper volume: P1 "Million-level extrapolation of coordination complexity of layered hybrid topology" (fan-in $O(N) \rightarrow O(\log N)$, verified extrapolation), P2 "BFT-lite stopping decisions," P3 "Fault governance RCT" (rework/mis-isolation), P9 "Quality saturation law and throughput curves" (cpu-proto), P10 "TransferBundle handoff completeness" (26 contract tests), P11 "Internal task market," P12 "Evidence grading" (LLM-as-judge bias discussion).
 2. UDOS v7.6.0 paper volume: P17 "Least-privilege orchestration" (scope granting), P33 "v7.6.0 master architecture."
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Evidence Discipline and Reproduction Notes

- This paper is an **evaluation methodology design paper**, no new experiments; two-layer aggregation formula (1), five-dimension weights, F23-a/b criteria are all design proposals, actual project-level scores [RESULT NEEDED] , forbidden to write as measured.
- EPOB five-dimension names and failure taxonomy categories have been verified; concrete constructions such as the "180-unit robustness package" are paper reports, not independently reviewed.
- If the two new Plan/Assignment labels use LLM post-hoc judging, note the judging bias P12 already discussed, prefer execution-style comparison.
- This paper does not claim EPOB has evaluated UDOS; the two are a complementary relationship of framework-centric and task-structure-centric.